

review for the final exam

1 The Final Exam

- Tuesday 5 May 2026, at 8am
- topics and guidelines

2 Some Questions

- Taguchi quality control
- linear programming
- cost benefit analysis
- machine learning
- ordinary differential equations
- root locus
- partial differential equations

MCS 472 Lecture 41
Industrial Math & Computation
Jan Verschelde, 24 April 2026

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The Final Exam

- Tuesday 5 May, from 8am till 10am, in 304 Burnham Hall.
- Or online, as the questions will be emailed to you.
- The exam is open book, open notes, with computer.
- The exam must be solved individually.
Submitting materials retrieved from the internet is plagiarism.
- Solutions must be in a Jupyter notebook with a Julia kernel.
- Answers must be submitted before or on 10am,
on Tuesday 5 May.
- Answering correctly five out of eight questions
gives a perfect score.
- Participation to the final is mandatory to pass the course.
- Submit to gradescope.

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topics on the final exam

Although the final exam covers everything, in this review we consider the most important problems.

In addition to this review:

- 0 Install all Julia packages we have used.
- 1 Review the homework problems.
- 2 Consider the review for the midterm exam.
- 3 Answers to the midterm exam are posted.

One of the questions on the final exam will be

Describe in one paragraph what you learned from another project presentation.

guidelines

common mistakes on submitted homework

Guidelines on submitting your answers:

- Submit one Jupyter notebook, no HTML, no PDF.
- Use section headings to mark the start of each question.
- Code cells are for text, markdown for explanations.
- Do not submit code cells that do not work.
- Include the plots whenever relevant.
- Submitting only code and computed results is not enough, add at least one sentence interpreting your results.
- Questions must be answered within the context of this course.
- Start submitting to gradescope *before* 10AM.

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Taguchi quality control

Suppose we measure the quality of watches as the deviation from the time at noon after one month without adjustments.

Watches which show a deviation of more than 3 minutes have to be discarded at a cost of \$10.

We have to choose between two ways to produce watches:

- 1 Watches produced by process *A* give after one month an average time of 12:01 with a standard deviation of 1 minute.
- 2 Watches produced by process *B* show 12:00 on average, but with a standard deviation of 2 minutes.

Which process is best? *A* or *B*?

- 1 Set up the quality loss function.
- 2 Use the expected loss to choose which process is best.

the quality loss function

The quality loss function is $L(X) = k(X - \theta)^2$ where

- X models the production,
- k is the loss coefficient,
- θ is the target value of the production,

has expected loss $E[L(X, \theta)] = k\sigma^2 + k(\mu - \theta)^2$, where

- μ is the mean value of the production,
- σ is the standard deviation of the production.

The target θ is 12:00.

- 1 $X = A$: $\mu_A = 12:01$, $\sigma_A = 1$ minute.
- 2 $X = B$: $\mu_B = 12:00$, $\sigma_B = 2$ minutes.

the loss coefficient

The quality loss coefficient k is the parameter in

$$L(x) = k(x - 12)^2.$$

The loss of a watch that deviates more than 3 minutes is \$10.

$$L(15) = \$10 \Rightarrow k3^2 = 10 \Rightarrow k = 10/9$$

Then the expected loss is $E[L(X, \theta)] = \frac{10}{9}\sigma^2 + \frac{10}{9}(\mu - \theta)^2$.

the expected loss for both processes

Then the expected loss is $E[L(X, \theta)] = \frac{10}{9}\sigma^2 + \frac{10}{9}(\mu - \theta)^2$.

① $X = A$: $\mu_A = 12:01$, $\sigma_A = 1$ minute.

$$E[L(A, 12:00)] = \frac{10}{9}1^2 + \frac{10}{9}(12:01 - 12:00)^2 = \frac{20}{9}$$

② $X = B$: $\mu_B = 12:00$, $\sigma_B = 2$ minutes.

$$E[L(B, 12:00)] = \frac{10}{9}2^2 + \frac{10}{9}(12:00 - 12:00)^2 = \frac{40}{9}$$

Because the expected loss for process B is higher than for A , process A is best.

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linear programming

A manufacturer produces two types of products, say A and B .

A sells at \$5 per unit, while B sells at \$7 per unit.

We have to decide how many units of each product to produce, with the aim of maximizing the profit, subject to constraints:

- ① We must produce at least 20 units of A , and offer at least 10 units of B .
- ② Our storage capacity is limited to 250 units.
- ③ We have only 600 hours of labor available,
 - ▶ one unit of A requires two hours of labor,
 - ▶ while one unit of B takes three hours of labor.

Do the following:

- ① Set up the linear programming model for this problem.
- ② Formulate the dual problem.

the objective function

Our objective is to maximize profit.

- Let x_A be the number of units we produce of A .
- Let x_B be the number of units we produce of B .

A sells at \$5 per unit, while B sells at \$7 per unit.

So we then arrive at the objective

$$\max 5x_A + 7x_B.$$

the constraints

- ① We must produce at least 20 units of A , and offer at least 10 units of B .

$$x_A \geq 20 \quad \text{and} \quad x_B \geq 10$$

- ② Our storage capacity is limited to 250 units.

$$x_A + x_B \leq 250$$

- ③ We have only 600 hours of labor available,
- ▶ one unit of A requires two hours of labor,
 - ▶ while one unit of B takes three hours of labor.

$$2x_A + 3x_B \leq 600$$

the maximization problem in standard form

$$\max 5x_A + 7x_B$$

subject to

$$-x_A \leq -20$$

$$-x_B \leq -10$$

$$x_A + x_B \leq 250$$

$$2x_A + 3x_B \leq 600$$

the dual problem

The dual of the maximization problem

$$\max 5x_A + 7x_B$$

subject to

$$-x_A \leq -20$$

$$-x_B \leq -10$$

$$x_A + x_B \leq 250$$

$$2x_A + 3x_B \leq 600$$

is the minimization problem

$$\min -20y_1 - 10y_2 + 250y_3 + 600y_4$$

subject to

$$-y_1 + y_3 + 2y_4 \geq 5$$

$$-y_2 + y_3 + 3y_4 \geq 7.$$

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cost benefit analysis

An investment of \$3,000 in information technology is expected to increase the monthly revenue by \$200.

Suppose the equipment has a life span of five years.

- 1 Use a discount rate of 3% to determine the payback period.
- 2 What is the return on investment?

the payback period

The payback period is the number of months after which the increased revenue surpasses the cost of the investment.

- 1 Compute the present value of the increase in revenue,
- 2 compounding at the monthly rate of $0.03/12$.

The present value $p(m)$ of the total increase in revenue at month m is

$$p(m) = \sum_{i=1}^m \left(\$200 e^{-i \cdot 0.03/12} \right),$$

where $\$200 e^{-i \cdot 0.03/12}$ is the present value of the \$200 of the i th month.

a Julia function

The present value $p(m)$ at month m is $p(m) = \sum_{i=1}^m (\$200e^{-i0.03/12})$.

In Julia:

```
julia> p(m) = sum([200*exp(-i*0.03/12) for i=1:m])  
p (generic function with 1 method)
```

```
julia> p(15)  
2940.76755767211
```

```
julia> p(16)  
3132.925445502574
```

We see that after 16 months, the present value of the increase in revenue has surpassed \$3,000, the cost of the investment.

the return on investment

The return on investment is the discount rate that yields zero savings.

- The cost of the investment is \$3,000.
- The life span is five years, or 60 months.

For the present value $p(m)$ at m months, we solve

$$p(60) = \$3,000,$$

where the discount rate d is a variable:

$$p(60) = \sum_{i=1}^{60} \left(\$200 e^{-i d/12} \right) = \$3,000.$$

using NLSolve.jl

Solve for d in $\sum_{i=1}^{60} \left(\$200e^{-i d/12} \right) - \$3,000 = 0.$

using NLSolve

```
"""
```

```
function f!(F, x)
```

defines the input to nlsolve.

```
"""
```

```
function f!(F, x)
```

```
    d = x[1]
```

```
    F[1] = 3000 - sum([200*exp(-i*d/12) for i=1:60])
```

```
end
```

```
roi = nlsolve(f!, [0.03])    # start at rate of 3%
```

```
println("ROI : ", roi)
```

the output

ROI : Results of Nonlinear Solver Algorithm

- * Algorithm: Trust-region with dogleg and autoscaling
- * Starting Point: [0.03]
- * Zero: [0.7574574772572242]
- * Inf-norm of residuals: 0.000000
- * Iterations: 9
- * Convergence: true
 - * $|x - x'| < 0.0e+00$: false
 - * $|f(x)| < 1.0e-08$: true
- * Function Calls (f): 10
- * Jacobian Calls (df/dx): 10

The interpretation of the * Zero: [0.7574574772572242]
is that the return on investment is 75.75%.

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machine learning

Consider the geographic prediction of the outcome of a referendum. Voters have three choices, labeled by

$y = -1$ (if disapprove), or $y = +1$ (if approve), or $y = 0$ (if abstain).

The pollsters collected 30 coordinates (x_1, x_2) , of 10 voters who disapproved, 10 who approved, and 10 who abstained.

With a neural network we want to geographically predict the outcome of the referendum.

- 1 Define the setup of the training data.
- 2 Draw the two-layer neural network for this prediction, using two neurons in the layer following the input layer.
- 3 What are the dimensions of the neural network?

the setup of the training data

This is a multiple classes classification problem.

The labels $y = -1, +1, 0$ are represented by the vectors

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{respectively.}$$

The training data consists of two matrices X and Y .

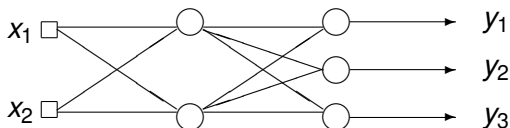
$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,30} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,30} \end{bmatrix},$$

where the k columns stores the coordinates of the k th voter.

The matrix Y is a 3-by-30 matrix, with in the k th column the vector encoding of the choice of the voter.

the two-layer neural network

Draw the two-layer neural network for this prediction, using two neurons in the layer following the input layer.



The weights and the bias vectors are

$$w_2 \in \mathbb{R}^{2 \times 2}, w_3 \in \mathbb{R}^{3 \times 2}, b_2 \in \mathbb{R}^2, b_3 \in \mathbb{R}^3,$$

so we have in total $4 + 6 + 2 + 3 = 15$ parameters to determine.

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ordinary differential equations

A crack in the hull of a ship at sea suddenly breaks open and the ship starts taking water in.

As the crack is widening, the rate at which the volume of water inside the ship increases is one tenth of the volume already in the ship.

The pump on board can pump out water at a rate of 0.5 m^3 per second.

When the pump starts pumping water out, there is already 20 m^3 of water inside the ship.

- 1 Set up the ODE to model the evolution of the volume of water inside the ship as time advances.
- 2 Only when the volume of water inside the ship becomes zero, one can cover the crack in the hull and save the ship.
Solve the ODE. Will the ship sink?

the ODE model

$V(t)$ is the volume of water inside the ship in function of time t .

Consider the facts:

- The rate at which $V(t)$ increases is one tenth of $V(t)$.
- The pump works at the rate of 0.5 m^3 per second.
- When the pump starts, there is already 20 m^3 water.

This leads to the initial value problem

$$\frac{dV}{dt} = \frac{1}{10} V(t) - 0.5, \quad V(0) = 20.$$

solving the ODE

using SymPy

```
t = Sym("t")
V = SymFunction("V")
ode = diff(V(t), t) - V(t)/10 + 1/2
sol = dsolve(ode, V(t), ics=Dict(V(0) => 20))
println(sol)
```

The output is

$\text{Eq}(V(t), 15.0 \cdot \exp(t/10) + 5.0)$

which gives $V(t) = 15e^{t/10} + 5$. The ship sinks.

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root locus

Consider the plant with transfer function

$$P(s) = \frac{5s^2 - 88s - 43}{-73s^2 + 25s + 4}.$$

- 1 Show that the plant is unstable.
- 2 Apply the root locus method to find a simple proportional feedback constant κ to stabilize the plant.

the plant is unstable

The roots of the denominator of

$$P(s) = \frac{5s^2 - 88s - 43}{-73s^2 + 25s + 4}$$

are

$$\frac{25}{146} - \frac{\sqrt{1793}}{146} \quad \text{and} \quad \frac{25}{146} + \frac{\sqrt{1793}}{146}.$$

As the (real part of the) second root is positive, the plant is unstable.

the root locus method

In the root locus method, we look at the roots of

$$(-73s^2 + 25s + 4) - \kappa(5s^2 - 88s - 43)$$

for increasing values of κ .

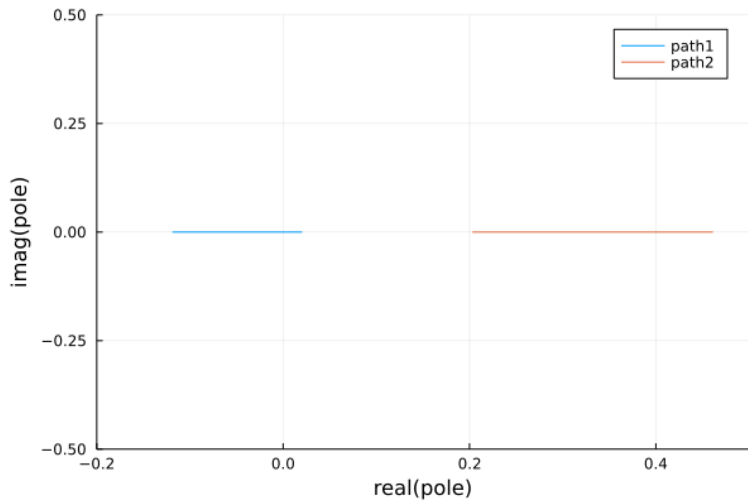
The κ is the parameter in the controller $H(s) = \kappa$.

The closed loop system has transfer function

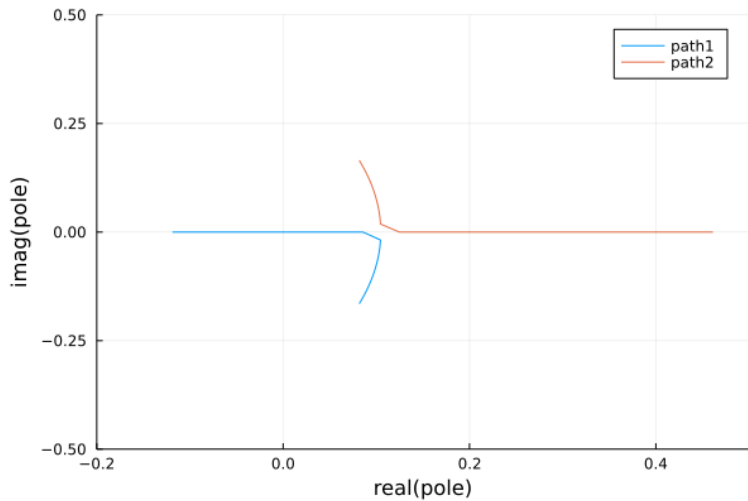
$$\frac{P}{1 + \kappa P}.$$

We apply the adjusted code of L-35.

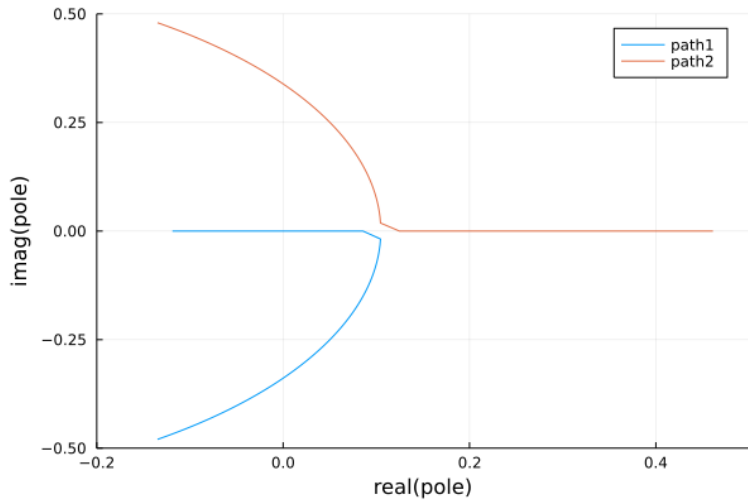
κ from 0 to 0.1



κ from 0 to 0.15



κ from 0 to 0.5



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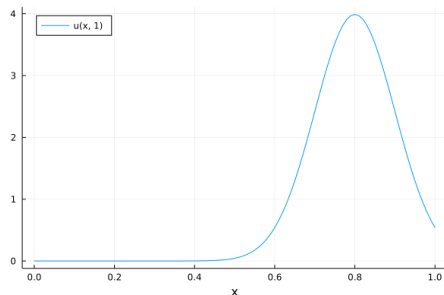
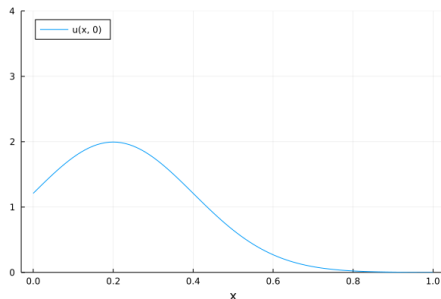
partial differential equations

Model the education process by a PDE.

- The function $u(x, t)$ models the distribution of the scores at time t .
 - For $s \in [0, 1]$, $\int_0^s u(x, t) dx$ gives the number of students with score s or less. The s can be interpreted as a percentage.
 - At $t = 0$, $u(x, t)$ obviously corresponds to the Bell curve which models the individual IQs.
 - As the semester progresses, the profile of the Bell curve changes.
- 1 What type of PDE would you choose to model the education process? Justify your choice.
 - 2 What are the critical parameters in your model?
How does one characterize the best education process?

the PDE model

Assuming bell shaped profiles of the scores:



$$u(x, t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

- At time $t = 0$: $\mu = 0.2$, $\sigma = 0.2$.
- At time $t = 1$: $\mu = 0.8$, $\sigma = 0.1$.

the transport equation with diffusion

The advection diffusion equation is

$$\frac{\partial u}{\partial t}(x, t) + \frac{\partial(vu)}{\partial x}(x, t) = k \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x).$$

Over time, we expect from the distribution of the scores:

- the mean μ shifts away from the low 0.2 to a higher value,
- the deviation σ shrinks from 0.2 to a lower value.

the parameters in the model

$$\frac{\partial u}{\partial t}(x, t) + \frac{\partial(vu)}{\partial x}(x, t) = k \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x).$$

The parameters in the model are

- (μ, σ) , the mean and deviation of the score distribution.
- (v, k) , the speed and diffusion (or better: concentration) constant.

The best education process is characterized by

- 1 A high value of v .
- 2 A strongly negative value of k .

Ideally, it does not take long to get up to speed,
and there is little variation in the capabilities of the students.